OMM PRAKASH PADHIARY

Systems & Hardware Engineer · AI/ML Developer · Full Stack

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Driven B.Tech Artificial Intelligence student at NIT Rourkela with a unique blend of frontend web development, machine learning, and systems engineering skills. Highly disciplined self-starter utilising intensive 14-day learning sprints to master complex topics like Deep Learning, ROS 2, and C++ hardware execution. Currently building real-time vision sensor models and seeking software engineering or AI/ML internship opportunities to bridge intelligent algorithms with hardware systems.

🎓 Education

National Institute of Technology (NIT) Rourkela

B.Tech in Artificial Intelligence | 2022 – Present

Rourkela, India

📚 Coursework

- Data Structures & Algorithms
- Math Foundations & Programming (Python, C)
- Machine Learning & Computer Vision

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Technical Skills

Languages

Python · C/C++ · JavaScript · HTML5 · CSS3/SASS · SQL

AI & ML

Deep Learning · Computer Vision · TensorFlow · Keras · Scikit-Learn · RAG · LangChain · Pandas · NumPy

Hardware & Systems

ROS 2 (Jazzy) · ESP32/STM32 · Real-time Systems · Cybersecurity Engineering

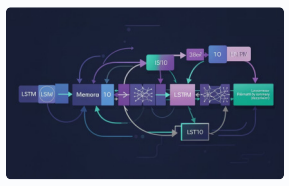
Web Development

React.js · Next.js · Tailwind CSS · Bootstrap · Responsive Design

Tools & Platforms

Git/GitHub · Linux/CLI · VS Code · Jupyter Notebook · Figma · Docker Basics

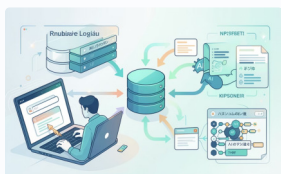
Technical Projects



Next-Word-Predictor

JAVASCRIPT · DEEP LEARNING · LSTM

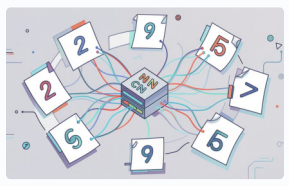
Architected a deep learning model utilising Long Short-Term Memory (LSTM) networks to execute sequence-based text prediction. Achieved **87% accuracy** on test datasets.



RAG-DSA-Project

JAVASCRIPT · RAG · NLP

Developed a Retrieval-Augmented Generation system tailored for querying and solving complex Data Structures and Algorithms concepts with **context-aware responses**.



Computer Vision: Number Recogniser

PYTHON · TENSORFLOW · CNN

Engineered a CNN-based computer vision application classifying handwritten digits with **95% accuracy** using raw pixel data processing.

House Price Predictor

PYTHON · SCIKIT-LEARN · JUPYTER

Trained and evaluated a machine learning regression model predicting housing prices using multiple feature datasets, achieving **92% R^2 score**.

DSA-AI Assistant

JAVASCRIPT · API INTEGRATION

Built an interactive AI-powered web assistant designed to guide developers through solving and understanding DSA problems with step-by-step explanations.

Mini-Cursor IDE Clone

Developed a lightweight, responsive browser-based clone of the Cursor AI IDE interface with functional editor features and dark theme styling.

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Continuous Learning & Methodology

A disciplined, sprint-based approach to mastering cutting-edge technologies — bridging intelligent algorithms with real-world hardware systems.

14-Day Sprint Cycles

Utilise structured 14-day learning sprints to rapidly acquire and implement new technical skills in AI/ML and systems engineering domains.

Vision Sensor Models

Building real-time vision sensor models that integrate deep learning outputs with live hardware sensor feeds for intelligent perception.

C++ Hardware Execution

Writing non-blocking, high-performance C++ code optimised for embedded hardware execution on microcontrollers like ESP32 and STM32.

ROS 2 Integration

Bridging deep learning outputs with real-time sensor processing pipelines using ROS 2 (Jazzy) for robotics and autonomous systems.

95%

CNN Accuracy

Digit recognition model

92%

R^2 Score

House price predictor

87%

LSTM Accuracy

Next-word predictor

14

Day Sprints

Rapid skill acquisition

Open To: Internships & Collaborations in AI/ML, Systems Engineering, and Full-Stack Development. Reach out at **prakashomm125@gmail.com** or connect on LinkedIn at **linkedin.com/in/ommparakash**.

> "The resume of a high-end AI Architect is not just a list of skills — it is a proof of systems thinking, disciplined learning, and the courage to bridge intelligence with hardware."

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